

Article

Long-Term Forecasting: A MAED Application for Sierra Leone's Electricity Demand (2023–2050)

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Abstract: Sierra Leone is an electricity-poor country with one of the lowest electricity consumption per capita rates across sub-Saharan Africa. Yet, with ambitious targets to transform and stimulate its economy in the coming decades, energy demand forecasting becomes an integral component of successful energy planning. Through applying the MAED-D (version 2.0.0) demand software, this research study aims to generate Sierra Leone's electricity demand forecasts from 2023 to 2050. Three novel scenarios (baseline-, high-, and low-demand) are developed based on socio-economic and technical parameters. The baseline scenario considers the current electricity sector as business-as-usual; the high-demand scenario examines an ambitious development future with increased economic diversification and mechanisation, and the low-demand scenario examines more reserved future development. The modelled scenario results project an increase in electricity demand ranging from 7.32 PJ and 12.23 PJ to 5.53 PJ for the baseline-, high-, and low-demand scenarios, respectively, by 2050. This paper provides a base set of best-available data needed to produce an electricity demand model for Sierra Leone which can be used as a capacity-building tool for in-country energy planning alongside further integration into data modelling pipelines.



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Keywords: demand forecasting; MAED; energy modelling; energy planning; capacity building; electricity; power sector

1. Introduction

1.1. Background

Sierra Leone is a lower-income agrarian economy situated in West Africa, bordered by Guinea, Liberia, and the Atlantic Ocean. In 2020, the African Development Bank ranked Sierra Leone 46th out of the 54 sub-Saharan African countries on the Africa Infrastructure Development Index and 44th on the Electricity Index [1]. Similarly, in the United Nations 2020 Human Development Index, it ranked 181st out of 191 [2]. Sierra Leone is one of the most electricity-poor countries throughout sub-Saharan Africa, with the country's energy needs remaining largely unserved [3]. The supply of electricity is historically intermittent and unreliable, with frequent electricity blackouts. Additionally, around 5 million people were without access to electricity in 2019 [4]. The electricity sector experiences a substantial demand and supply gap, plagued by distribution losses reaching up to 50% [5]. Results from the 2018 household survey indicated that poverty is still high across Sierra Leone, particularly in rural areas, with 10.8% of the population living in extreme poverty and 57% in poverty [6].

Sierra Leone's energy system advancement and development was severely impacted by national events, including the Ebola disease outbreak [7]. Additionally, global events

such as the COVID-19 pandemic, the rapid iron ore price decline, and the Russian invasion of Ukraine are further inhibiting energy system supply, investments, and clean energy technological expansion [8,9]. These external shocks remain key barriers experienced by Sierra Leone in its electrification and energy advancements efforts. Sierra Leone has a series of ambitious and thorough development goals, spearheaded by its 'Vision 2039 statement', which aspires to gain middle-income country status by 2039, launched alongside the latest medium-term national development plan 2024–2030 [10]. The main goals include accelerating agricultural development, expanding free education, increasing safe water access, and diversifying the economy. Energy sector goals include providing reliable and affordable power through consolidation, improvement, and expansion [11]. To achieve this, a domestic budget allocation of 15.6 million US dollars (USD) was assigned to distribution network improvements, plant upgrades, and increased electricity generation alongside working with international partners and foreign direct investors.

Modelling software can be used by policy makers to assess the impacts of different scenarios on energy systems, supporting planning and decision making. Mid- to long-term planning is crucial to allow for the allocation of resources and securing of investments for developing countries [12]. Demand forms an integral foundation of energy planning and insights into possible projections can aid in policy creation, yet access to data is often a barrier to utilising energy demand modelling to support such decision making [13]. This study will collect the 'best-available' social, economic, and technical data and utilise a quantitative simulation modelling methodology to produce electricity demand forecasts for Sierra Leone's electricity sector. The data collected and the model developed can be adopted and developed by in-country planners, academics, and policy makers for further exploration. The demand projections produced can further be used to aid in energy-planning processes, such as capacity expansion, funding exploration, and policy development. A novel application of the Model for the Analysis of Energy Demand (MAED) software (version 2.0.0) for Sierra Leone's electricity sector is therefore explored to gain a critical insight into potential future electricity demand projections under various scenario parameters, overcoming historic barriers experienced by energy planners in Sierra Leone [14].

1.2. Literature Review

Energy modelling has developed greatly as a methodology to aid energy analysis, planning, and policy development across the last few decades [15,16]. Modelling tools have been used to make sure energy sector development aligns with sectoral needs through optimising and/or simulating demand and supply to produce cost-efficient energy profiles [17]. Subsequently, energy modelling as a methodology can critically aid in energy-planning processes by producing evidence-based, scientific analyses to support policy pathway development and investment decisions [18]. One application of energy modelling software is studies producing energy sector demand projections. Verwiebe et al. [19] conclude that well-founded energy demand forecasting is one of the most crucial components of energy planning and modelling, having a significant impact on present and future energy system decisions. This is particularly the case for long-term forecasting, with consequences affecting costly expansion and planning decisions and with major impacts in cases of over- or underestimation.

Energy demand forecasting as a field of energy planning has seen the development of various methodologies, applying different approaches and techniques, which have been extensively applied in academia [20,21]. The methodological scope is large and widely contested, and it includes artificial neural networks, fuzzy logic, deep learning, econometrics, and time-series models [19]. Each methodology brings with it a different starting set of assumptions, data requirements, and subsequent projections produced. Ghalekhondabi et al. [22] categorise demand methodologies into two main typologies: causal, such as artificial neural networks and regression models, or historical, including grey prediction and time-series models. In causal methodologies, the output of energy consumption is seen to have a direct

cause-and-effect relationship with different economic, social, and environmental input factors. Alternatively, historical methods use previous variable values to produce future forecasts. Bhattacharyya et al. [14] typologise two categories of models, namely simple and sophisticated approaches. Simple approaches require minimal data and skills, whereas sophisticated models adopt a more advanced methodology to produce forecasts with higher validity and explanatory power. Traditionally, neural networks have been the dominant demand methodology, yet their computational time is extensive, relying on advanced skill and knowledge [19]. Contemporary studies are not dominated by such methodologies, instead seeing a rise in the application of econometric, end-use, and hybrid models [20]. Despite the breadth of approaches to forecasting demand, studies have concluded that none of the methodologies outperform the others, and there is no consensus amongst scholars on the most successful approach to apply [22].

A historic barrier to the application of advanced demand methodologies is that they are often time-intensive, heavily data-driven, and rely on advanced skills and knowledge to be able to successfully produce models [14]. This is a particularly pressing barrier for energy management and sector development planning in many lower- and middle-income (LMIC) economies, where energy planning units are often scattered, with limited capacity and a lack of experience or skills with advanced modelling [21]. Further, data availability is restricted and intermittent, with national data sets often experiencing poor quality control and a lack of consistency with data collection methodology, and international data sets often relying heavily on assumptions to account for a lack of high-resolution data [13]. Additionally, Bhattacharyya et al. [14] challenge the assumption that methodologies traditionally applied to developed economies can be similarly translated to developing countries. They state that developing countries have unique socio-economic features, such as large informal sectors, high presence of inequality, and an urban–rural divide, alongside an active transformation in lifestyles and the economy. Alongside this, developing countries' energy sectors are often categorised by a high reliance on traditional fuels, low efficiency, high transmission and distribution losses, and supply shortages. Such features are not identified in developed countries and have not been considered using traditional demand-forecasting methodologies [22]. Consequently, this study attempts to overcome these barriers by applying a generic, or simple, modelling approach [13].

Previous studies have explored demand forecasting at a regional level through examining power pool electricity projections. Adeoye and Spataru [23] use MATLAB to produce hourly electricity forecasts across the West Africa Power Pool (WAPP) from 2016 to 2030. They develop their own hybrid methodology, which employs a bottom-up approach for households and a top-down approach for industrial, commercial, and services demand projections. Similarly, Ouedraogo [24] adopt a bottom-up approach through LEAP software application to produce regional power pool electricity projections across Africa until 2040 across baseline, renewable energy, and energy efficiency scenarios. Finally, Semekonawo and Kam [25] compare ARIMA and linear regression methodologies to assess their suitability in producing electricity demand forecasts for countries in West Africa. Whilst such studies have included Sierra Leone within their projections, they often lack the spatial and temporal resolution to produce meaningful and useful national-level outputs for key actors within Sierra Leone to adopt and employ within research and policy making.

There is currently only one published contemporary study which models national-level energy demand projections in Sierra Leone. Conteh et al. [26] use Long-range Energy Alternatives Planning System (LEAP) software to produce national electricity demand projections from 2019 to 2040 across three modelled scenarios, including a baseline-, middle-demand, and high-demand scenario. The study not only considers the demand forecasts produced under each scenario's parameters, but also examines the transformation and supply side required to meet the demand scenarios and the associated CO₂ emissions produced. The study finds that their baseline energy projections increase from 791.1 GWh in 2019 to 1812.5 GWh in 2040, at a total increase of 129% over the modelling period. This increases to 144% in the middle- and 233% in the high-demand scenarios. The study's

projections are constructed from a base year of five years ago. However, recent macroeconomic events such as the global COVID-19 pandemic and the Russian invasion of Ukraine have significantly impacted electricity consumption patterns [8], requiring exploration into updated projections in response to such changes. The LEAP study also has limited transparency surrounding its data input sources and it is often unclear where such figures were acquired. Whilst LEAP is free to download, the model is not open-sourced and therefore lacks the transparency and open collaboration which open-sourced software facilitates [13]. Subsequently, there is a need to build on previous research, applying updated data to a comparative open-sourced demand simulation tool, such as the Model for the Analysis of Energy Demand (MAED), expanding the base modelling years.

Previously published studies, outlined in Table 1, have applied the MAED to produce projections at both a national and sub-national level with case studies across the world. Recent MAED applications have seen an increase in published studies focusing on sub-Saharan Africa. Kanté et al. [27] used the MAED to assess various electrification scenarios for the region of Taoussa in Mali from 2020 to 2035. The scenarios employed within the study integrated considerations of demographic, economic, technological, and lifestyle evolution including food security, industrial mechanisation, poverty eradication, and IT equipment expansion. Mpholo et al. [28] applied the MAED to produce electricity demand forecasts for Lesotho from 2010 to 2030, focusing on national electrification targets within their scenario creation and analysis. Kichonge et al. [29] similarly used the MAED to obtain forecasts for Tanzania’s whole energy system, including the transport sector, from 2010 to 2040 across three modelled scenarios.

Table 1. Related works on energy demand modelling with MAED application.

Author	Year	Case Study	Model Period	Level	Scenarios	Ref.
Hainoun et al.	2006	Syria	1999–2030	National	3	[30]
Jovanovic et al.	2009	Serbia	2005–2020	Local	5	[31]
Kichonge et al.	2014	Tanzania	2010–2040	National	3	[29]
Nakarmi et al.	2014	Nepal	2010–2030	National	4	[32]
Guemene Dountio et al.	2016	Cameroon	2012–2035	National	3	[33]
Sieed et al.	2020	Bangladesh	2020–2050	National	5	[34]
Kante et al.	2021	Mali	2020–2035	Local	3	[27]
Mpholo et al.	2021	Lesotho	2010–2030	National	3	[28]
Liun et al.	2022	Indonesia	2020–2060	National	1	[35]
Battulga and Dhakal	2023	Mongolia	2020–2050	Local	4	[36]

Their study examined a series of policy and behavioural changes across scenarios including urbanisation, unexpected political and economic crises, lifestyle changes, and improved efficiency. Whilst MAED is a widely applied methodology globally, the model’s development and application for Sierra Leone’s energy sector remains unexplored. Therefore, this study builds on previous demand studies, with a novel application of Sierra Leone’s electricity sector to the MAED simulation software, extending the modelling period to 2050 and updating socio-economic and technical inputted data based on best-available national and international data sets.

2. Materials and Methods

2.1. Methodology

The Model for the Analysis of Energy Demand (MAED) was chosen as it is a flexible, free, and open-sourced scenario-based tool which produces disaggregated annual energy demand projections. The MAED is a generic, or ‘simple’, model created by the International Atomic Energy Agency (IAEA) and employs a bottom-up simulation methodology [14]. The MAED is an internationally renowned demand projection model and has been actively used by governments, energy planners, and academics to produce demand projections in over 135 countries [37]. The MAED-D, previously titled MAED-2, offers a dynamic model in which the user can evaluate the energy system at varying geographical and sector

aggregation scales [34]. It combines energy intensities across sub-sectors per energy type (specific electricity use, thermal, motive) and per fuel type (electricity, fossil fuels, traditional fuels, biomass, solar thermal) with driving factors (GDP growth rates, population growth, efficiencies, etc.) to derive the final demand projections. The model reduces demand outputs for the energy system into four end-use sectors, namely industry, transport, services, and households, and produces both useful and final energy demand per sector and per fuel type [32]. A more in-depth overview of the methodology employed within the software can be found in the published handbook [38].

The rationale behind utilising the MAED methodology lies in its ability to overcome barriers specifically experienced by developing countries in their energy planning and management [14,21]. Previous MAED applications have demonstrated the model's usefulness for developing countries due to its flexible and dynamic framework, allowing the user to self-define the sectors, sub-sectors, end-use energy services, and fuels which are relevant to their case study [36]. The user can produce a disaggregated whole energy system, selecting which sectors to represent and at a resolution best fit for the data available in their chosen case study [28]. In cases where data availability is high, the MAED can be manipulated to produce detailed energy demand, with a higher resolution and sectoral disaggregation. Alternatively, in cases where data availability is limited, the MAED can be aggregated to respond to this, increasing the model's applicability and accessibility to a wider range of potential users. The model only requires the collection of base-year data, which overcomes barriers experienced by cases with limited and intermittent data and by planning units with restricted resources and time allocation [35]. The MAED also accounts for the substantial rural and urban divide in overall energy usage and for the energy fuel mixes often featured in developing countries, with a predefined disaggregation of the household sector into urban and rural and allowing the user to further define different types of dwellings [29].

Additionally, the MAED software application has a strong capacity building potential. The Climate Compatible Growth (CCG) programme, in collaboration with the Open University and partner organisations, have produced and released a series of 13 free courses and training material on energy modelling software and planning, including the MAED [39]. The MAED course aids users in the process of creating a basic whole-energy-system demand forecast for the industry, services, household, and transport sectors. Alongside this, the CCG offers model development and troubleshooting support through online community channels and in-person energy modelling platforms which occur three times a year [13]. The MAED is a simple and accessible model which is suitable for beginner modellers and does not require any previous modelling experience or advanced skillsets. The MAED is free to download and open-source, with an easy-to-use interface, and does not require advanced or expensive technology to run, helping to reduce barriers to model development and uptake in countries with low capacity [38]. This is particularly useful for developing countries, where human and technological resources are typically constrained and there is a lack of energy-planning units [13].

Energy-modelling capacity within the Ministry of Energy's planning unit has been a historic barrier to the creation and integration of modelling into planning processes within Sierra Leone. Previous energy modelling to aid planning decisions has been outsourced to international consultants from private firms such as AF-Mercados, who created the Integrated Resource Plan in December 2020, in collaboration with the Millennium Challenge Coordinating unit [40]. As modelling and analysis have historically been outsourced beyond Sierra Leone's planning unit, the in-country capacity to uptake, review, develop, and produce their own models have been significantly reduced [41]. Additionally, accurate and up-to-date data availability remains a barrier to energy model development and analysis [21]. The data needed to successfully run energy models are time-intensive to collect and are often inconsistent, inaccurate, and not open-sourced [13]. This study aims to overcome some of the historic barriers to energy modelling within Sierra Leone, producing

an open-source, accessible, and replicable baseline-demand forecast and starter set of data which in-country modellers, analysts and planners can adopt and develop.

This study looks at the electricity demand in Sierra Leone for the end-use sectors of industry (agriculture and mining), services, and households (urban and rural). The modelling period of 2023 to 2050 is chosen to provide a long-term forecast and the years 2018 to 2023 are used to create a baseline from historic demand data. Subsequently, this study examines the research question of ‘how do different social, economic, and technological parameters impact future electricity demand in Sierra Leone?’.

2.2. Data

A full overview of the data included within this model and the subsequent sources it was acquired from is outlined in Kiley et al. (2024) [42] and the Supplementary Files of this paper. The data included within the model were collected via an extensive literature review of databases, reports, and journal articles [43]. Data discrepancies across the literature were present, and subsequently, national data sets were prioritised over international ones to form a set of ‘best-available data’ on the assumption that national data sets were more likely to hold the most up-to-date and accurate data. The data outlined form the baseline scenario which simulates a business-as-usual possible future for Sierra Leone’s electricity demand forecast. The units used within the MAED were chosen to be peta-joules (PJ) to allow for the seamless integration of results into optimisation-modelling software as demand input, creating a data-modelling pipeline [44].

2.2.1. Demography

Sierra Leone’s population in 2018 totalled over 7.8 million people, having almost doubled over the previous 15 years due to a sharp increase in population growth [45]. Most of the population reside in the Eastern and Southern regions; however, the Western region has the highest population density, with a substantial percentage of the population residing in the capital of Freetown, which is currently home to over 1.2 million people [46]. Sierra Leone has a young population, with the 2015 census showing that over 40% of the total population was under the age of 15 and over 80% were under 35 [47]. Additionally, the census revealed that the proportion of economically active population at working age had fallen to just over 50% due to rising levels of unemployment and poverty. Most of the labour force work in the industry sector, primarily in agriculture, yet there has been a growth in the number of workers in the services sector. The distribution of labour force by sector varies by region, with the Western region consisting of mostly service workers with a very small agrarian workforce [48]. The demographic information collected and included in the MAED model is outlined in Table A1, along with a full list of references.

2.2.2. GDP

The country’s total GDP in 2018 reached over USD 4.09 billion, situating it as a low-income economy [6]. Sierra Leone’s economy is undiversified and disproportionate in productivity, primarily relying on agriculture, which contributed to 57.4% of annual gross domestic product (GDP) in 2018, followed by services at 36.7% and mining at 5.9%. Historically, Sierra Leone’s GDP growth has been inconsistent and unpredictable; for example, it changed from −20.6% in 2015 to 6.1% in 2016 due to decades of uneven economic growth across sectors, post-war economic recovery, and a series of external shocks and crises. This trend continued across the base-year modelling period of 2018–2023, with GDP growth rates ranging from −2% to 5.25%. Due to the undiversified and small economy, a key policy priority remains to diversify and transform the economy to promote growth and build resilience to shocks [49].

Sierra Leone’s economy has historically relied on its agricultural sector, primarily producing rice, sugar, oil palm, and cocoa crops. Agricultural practices across the country are traditional, primarily in small-scale farms; consequently, agriculture is not an energy-intensive sector [50]. Sierra Leone also has a substantial mining sector, primarily

formed of diamond, gold, and iron ore extraction, with mineral exports contributing over 90% of total export revenue [51]. Recent years has seen a realignment of labour at a sectoral level, with a growth in employment shifts from agriculture to services, yet this has not yet been reflected in sectoral GDP [49]. Additionally, there have been initial efforts to ramp up mining productivity through increased foreign direct investment, yet the economic benefits of this have not yet been achieved. The GDP information collected and included in the MAED model is outlined in Table A2, along with a full list of references.

2.2.3. Electrification

Sierra Leone's electrification programme remains historically underdeveloped. In 2018, national electricity access rates were 22.4% of the total population, yet distribution remains unequal and primarily focused in urban areas, with a 53.2% penetration, compared to just 6.4% in rural areas [26,52,53]. The grid-connected electricity system is concentrated in the Western region, with over 80% of the country's consumption, primarily in the capital city of Freetown, and in import and provincial hotspots [54]. Initial national electrification targets, outlined in the SE4ALL-backed Country Action Agenda, aimed to reach 92% national access by 2030, yet due to insufficient progress, the government of Sierra Leone has reevaluated the targets to reach 98% urban and 56% rural by 2040 [26]. The electrification rates included in this study are outlined in Table A3.

2.2.4. Historic Electricity Demand

The electricity demand in 2023 reached 1.183 PJ and was supplied via fossil fuels, hydropower, and solar power [42]. The electricity supply is divided into three main sectors, namely industry (split into agriculture and mining), services, and households (split into urban and rural). Historic electricity demand data were taken from the Energy Distribution and Supply Authority's (EDSA) historic generation data and adjusted to account for transmission and distribution losses [42]. Electricity demand in 2023 totalled 1.183 PJ, of which 53% was from services, 32% from households, 13% from mining, and 2% from agriculture. Figure 1 shows the total historic demand across the base years of 2018 to 2023 with a sectoral breakdown. The total sectoral demands (PJs) per year are outlined in Table A4.

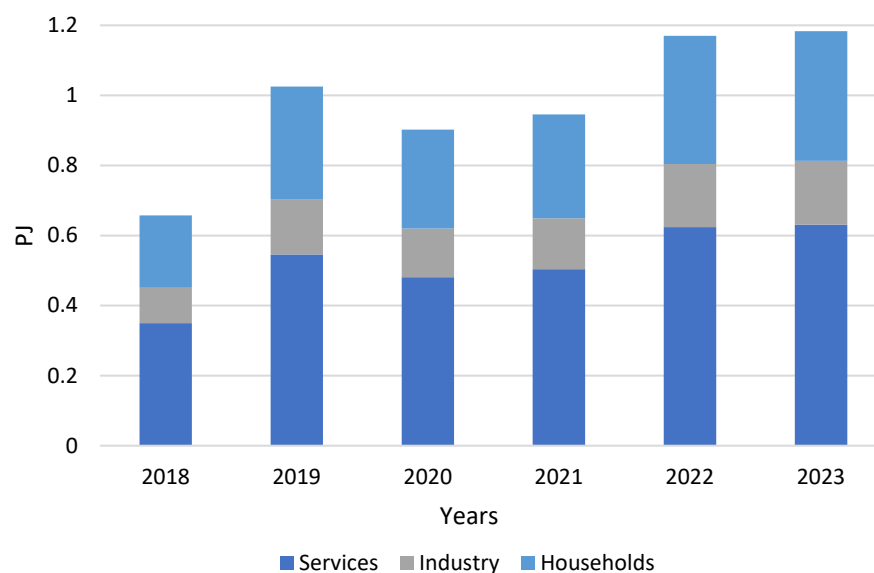


Figure 1. Historic electricity demand per sector from 2018 to 2023.

2.2.5. Energy Intensities

Some academics have proposed the decoupling paradigm, which states the need to separate economic growth from increasing energy demand [55]. However, this approach has been shown to be more relevant and applicable to cases of developed economies,

where energy efficiency and decarbonisation measures are much more advanced than in developing countries [56]. Research into energy planning and demand modelling for developing countries has continued to recognise the reliability of socio-economic indicators as an indicator of energy demand [14,56]. Consequently, this study assumes a relationship between energy demand and socio-economic development. Previous demand modelling for Sierra Leone has assumed a blanket base energy intensity increase of 3.5% annually [26]. This study assumes a varying increase in energy intensity depending on the sector, in line with historical electricity sector experiences and national development mechanisation and stimulation goals. Consequently, the energy intensity modelled for the services sector remains the same, that of the industry sector increases by 3.5%, that of urban households increases by 3% and that of rural households by 2% annually across the modelling period of 2023 to 2050. The energy intensities per sector are shown in Table A5.

2.3. Scenarios

Three scenarios—baseline-, high-, and low-demand—were developed and included within this study. The scenarios were chosen to provide a range within which future energy demand projections could sit based on the socio-economic and technical constraints identified. This aligns with previous demand modelling research [28,29,34] and allows for the integration of the produced demand-modelling results into a data modelling pipeline, providing a bandwidth of possible future demand to be integrated into by robust decision making (RDM) procedures when assessing energy expansion optimisation and uncertainty [37,44].

2.3.1. Baseline Demand

The baseline-demand scenario simulates a business-as-usual possible future in which Sierra Leone's economic and demographic parameters continue to increase at the 2023 rate. Additionally, the scenario assumes electrification targets, as outlined in Section 2.2.3, will be achieved. A full outline of the GDP and population growth rate assumptions across the scenarios can be found in Table A6.

2.3.2. High Demand

The high-demand scenario simulates a possible future in which Sierra Leone's national development goals of diversifying and promoting economic growth [10,49] are achieved. This is modelled through a continued increase in GDP growth from 2.75% in 2023 to 5.19% in 2025, where it remains until 2050, in line with Sierra Leone's Vision 2039 goal of reaching middle-income status. Alongside this, a diversification of the economy is achieved by increasing the share of the mining sector's GDP contribution to 20% and the services sector's to 40% by 2030 [10]. Additionally, the high-demand scenario assumes an increase in the annual population growth rate from 2023 to reach an annual rate of 2.54% [26]. An overview of the sectoral share changes per year can be obtained from Table A7.

2.3.3. Low Demand

The low-demand scenario simulates a possible future in which Sierra Leone national development is more reserved. National development goals of increased GDP stimulation and diversification are not achieved, and a reduction in GDP and population growth rates compared to the baseline scenario are experienced. The low-demand scenario assumes that the GDP growth rate remains at the 2023 rate of 2.75% until 2050. The low-population-growth scenario assumes a 0.03% reduction in growth rate from the previous year, in line with growth rate reductions from 2019 to 2020 and from 2020 to 2021.

3. Results

Under all potential future scenarios included, electricity demand is expected to increase significantly across the modelling period. Figure 2 illustrates the forecast for the total annual electricity demand for the three modelled scenarios. The baseline-demand forecast

increases from 1.18 PJ in 2023 to 7.32 PJ in 2050. This compares to a projected increase to 5.53 PJ in the low-demand case and to 12.23 PJ in the high-demand case. Figures 3–5 display the sectoral split of the final electricity demand across the three scenarios.

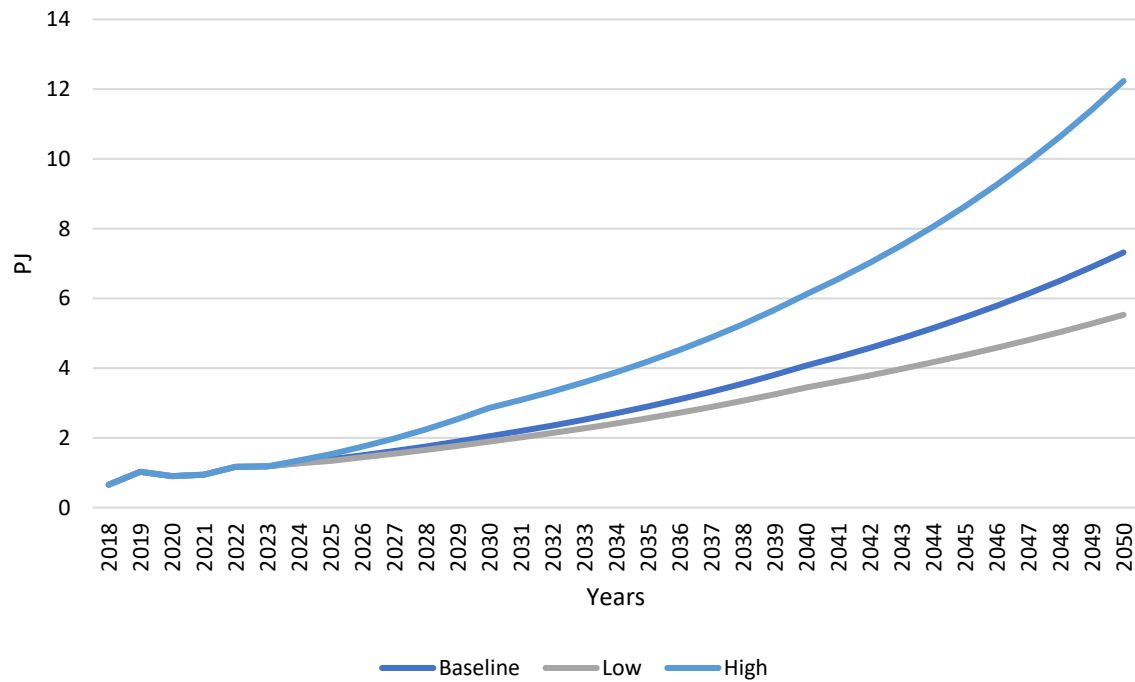


Figure 2. Total demand forecast from 2018 to 2050 across the three modelled scenarios.

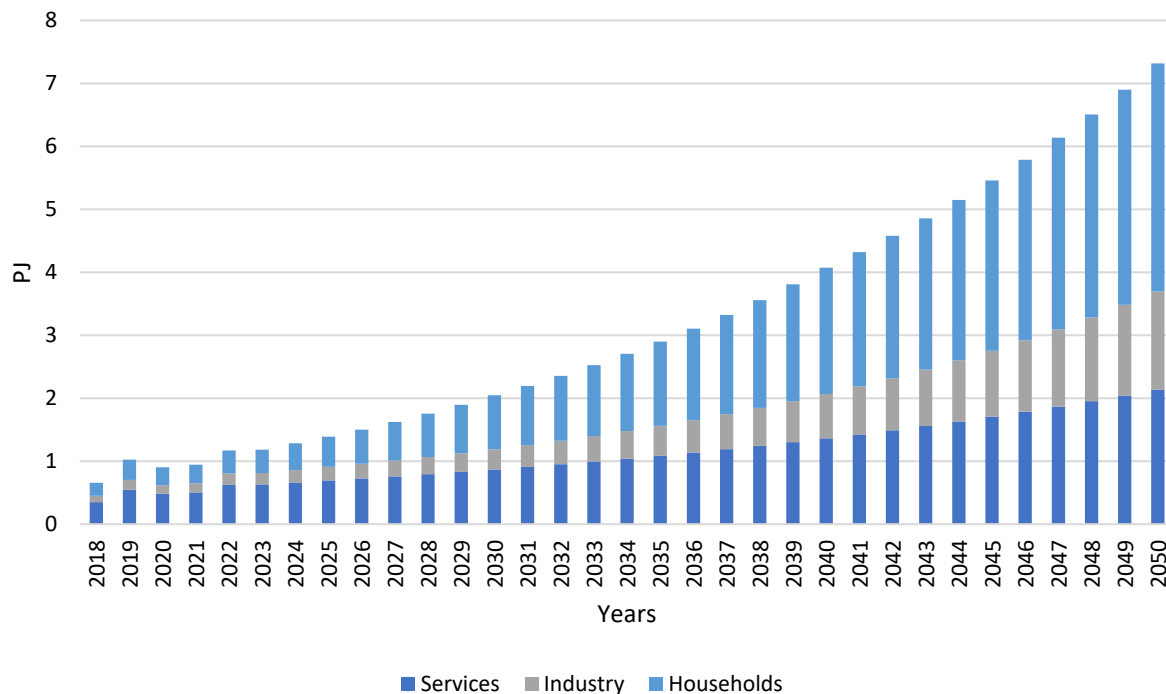


Figure 3. Baseline demand projection from 2018 to 2050.

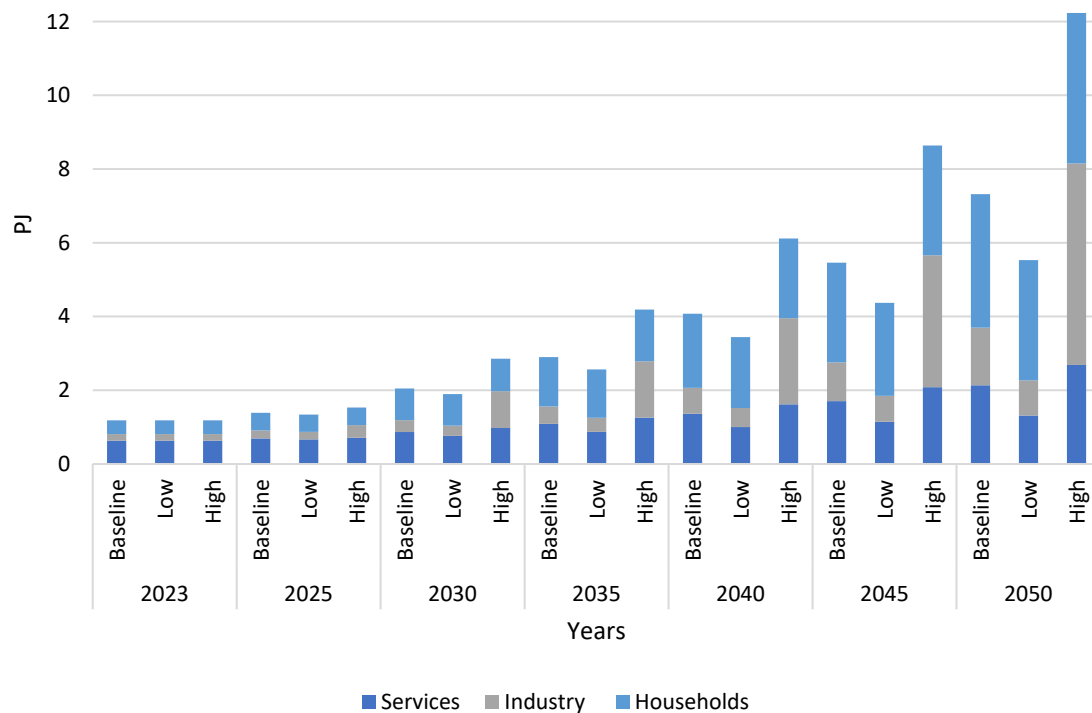


Figure 4. Sectoral demand projections across the three modelled scenarios from 2023 to 2050.

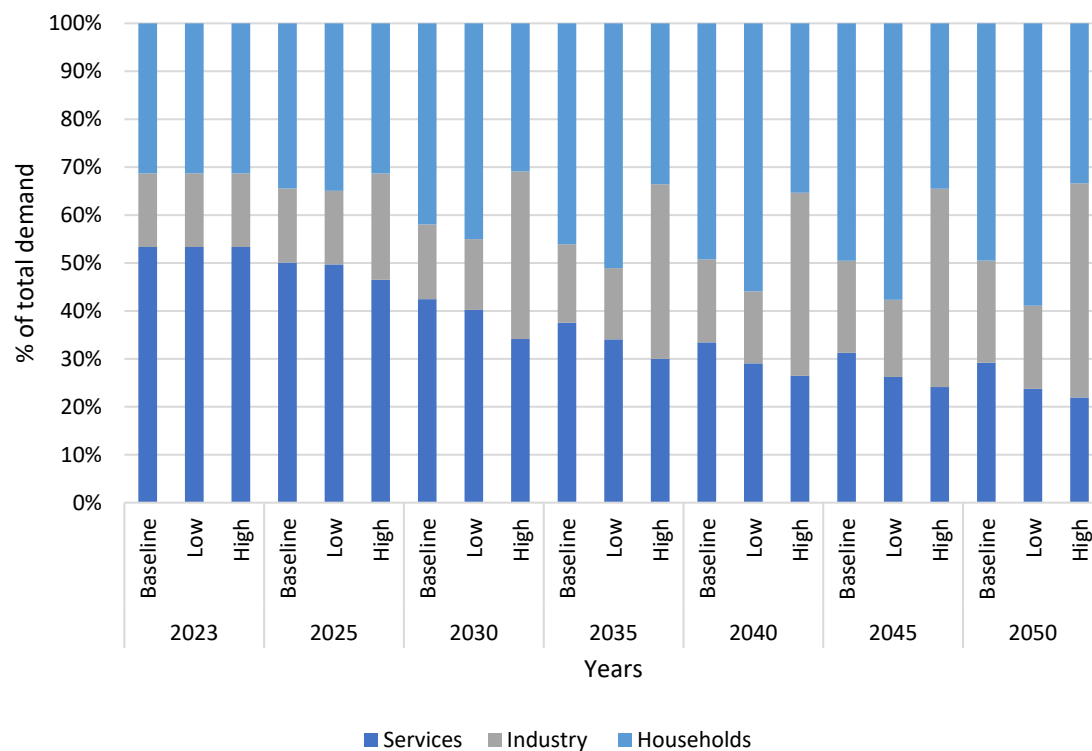


Figure 5. Sectoral split of total demand across the three modelled scenarios from 2023 to 2050.

In the baseline case study, the services sector forms the greatest share of demand at the beginning of the modelling period, followed by households and services, until 2030, where the household sector overtakes services, and its share of the total energy demand continues to increase until 2050. In 2050, the household sector forms 49.9% of the total demand, compared to 31.3% in 2023. This is followed by the services sector, at 29.2% in

2050 compared to 53.3% in 2023, and the industry sector, at 21.3% in 2050 compared to 15.4% in 2023. The low-demand scenario produces a demand split trend most closely related to the baseline scenario, where the services sector's share of the total demand is overtaken by households in 2029, and their share continues to increase for the remainder of the modelling period. In 2050, the household sector forms 58.9% of the total demand, followed by services at 23.7% and industry at 17.3%. The high-demand scenario produces a final demand mix which differs from the other scenarios, with the industry sector forming the highest share in 2050, followed by households and services, respectively. In the high-demand scenario, the industry sector overtakes services in 2030, followed by the household sector in 2033. By 2050, sectoral demand consists of 44.7% industry, 33.4% households, and 21.9% services.

4. Discussion

Three scenarios were developed following existing national demographic, economic, and technological goals alongside historic trends and data using the MAED-D energy demand model. The total final energy demand in 2050 was forecasted at 7.32 PJ, 5.53 PJ, and 12.23 PJ for the baseline-, low-, and high-demand scenarios, respectively, compared to 1.18 PJ in 2023. The demand forecasts produced show that in all scenarios, there will be a significant increase in electricity demand experienced across the modelling period. The total demand produced varies across the scenarios, with the high-demand scenario totalling 1.7 times and the low-demand scenario totalling 0.75 times the baseline demand in 2050. Whilst the scenarios produce a significant increase in demand, the final 2050 demand is still significantly smaller than other electricity sectors across sub-Saharan Africa. Additionally, Sierra Leone is still a developing economy with a huge potential for growth in the future and a significant increase in energy demand, particularly electricity, will be experienced to meet the ambitious economic stimulation and mechanisation goals to reach middle-income country status.

Across the scenarios, the household sector had the highest increase in percentage share in the low-demand scenario, from 31.3% in 2023 to 58.9% in 2050, at a difference of 27.6%. Similarly, the household sector also increased greatly in the baseline scenario from 31.3% in 2023 to 49.9% in 2050, a difference of 18.6%. Comparatively, the share of the household sector in the high-demand scenario remained similar across the modelling period, from 31.3% in 2023 to 33.4%, with only a range of 2.1%. The significant increase in the household sector's share can be accounted for by Sierra Leone's ambitious electrification targets. The household sector also forms a substantial share of the total electricity demand in 2023, at 31.3%, with an urban household electrification rate of 57% and a rural rate of only 4.9%. To reach 100% urban and 86% rural electrification in 2050 alongside an increase in annual consumption per household, there will need to be a considerable increase in demand.

The industry sector's share had the greatest growth in the high-demand scenario, increasing from 15.4% in 2023 to 44.7% in 2050, a difference of 29.3. This sizeable increase in industry sector demand is not unexpected, as the high-demand scenario considers a future in which the mining sector's GDP increases from 5.9% in 2023 to 20% in 2030 alongside an annual total GDP growth of 5.19%. Further, the mining sector is already the most energy-intensive sector within Sierra Leone, accounting for this significant increase in demand. In comparison, the industry sector's percentage share also increases in the remaining scenarios, reaching 17.3% in the low- and 21.3% in the baseline-demand scenario, but to a much less significant level. Finally, the services sector's total share decreases in all the modelled scenarios, from 53.3% in 2023 to 29.2% in the baseline-, 21.9% in the high-, and 23.7% in the low-demand scenario, to reach a total range of 31.4%. The total differences in percentage share by sector in 2023 compared to 2050 across the scenarios are shown in Figure 6.

The results projected in this study vary compared to previously published studies. In Conteh et al.'s LEAP study [26], Sierra Leone's baseline electricity demand is projected to reach 1812.5 GWh, or 6.53 PJ, in 2040. This compares to the baseline projection of 4.07 PJ produced in this study. The difference in the produced results is not surprising, however,

considering the disparity in the base-year electricity demand figures of 791.1 GWh, or 2.85 PJ, in 2019 in the LEAP study compared to the 1.023 PJ identified in this study. Alongside this, differing socio-economic and technical assumptions and parameters identified, and the alternative methodology applied, could explain the differences in the projections produced.

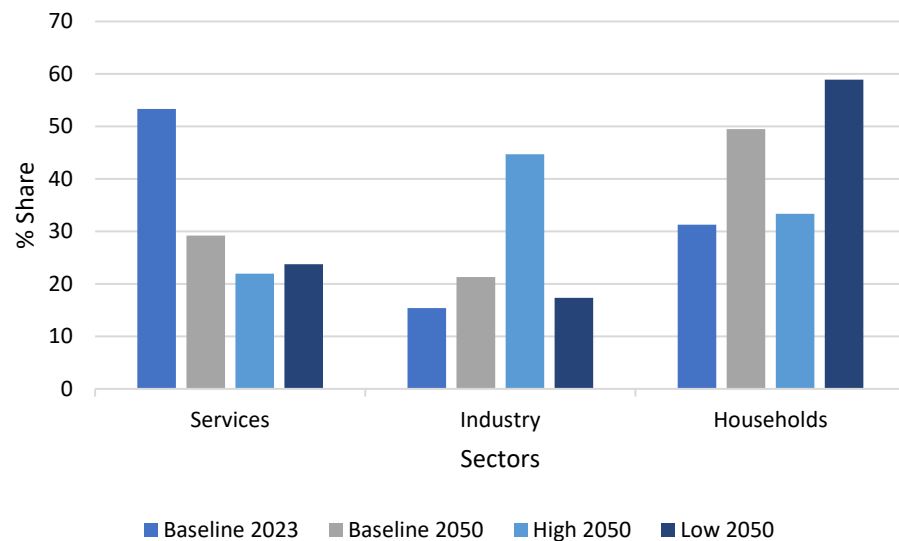


Figure 6. Percentage share difference by sector between the 2023 baseline and the modelled scenarios in 2050.

5. Conclusions and Research Recommendations

5.1. Future Research

This study was conducted to create an open-source starter set of ‘best-available’ social, economic, and technical data needed to produce a demand-forecasting model for Sierra Leone’s electricity sector utilising the MAED-D software (version 2.0.0). The findings from this study can be integrated into a data-modelling pipeline, such as Data-to-Deal [44], to allow for supply side optimisation, uncertainty analysis through robust decision making (RDM), and financing pathway identification. This will involve transferring the demand forecasts produced as an output in this study into an input for optimisation models such as the Open-Sourced Energy Modelling System (OSeMOSYS) [37]. As the scope of this research study was limited to the electricity sector, there is scope for further expansion to model Sierra Leone’s whole energy system, including thermal and motive power, alongside integrating the transport sector. The MAED software also provides an additional module, MAED-EL, which can be used to convert the yearly electricity demand produced in this study into hourly electricity consumption. Insights into load, flexibility, inertia, and reserve provision could be studied through MAED-EL’s application in further studies.

5.2. Limitations

Although the methodology employed in this study addresses the historical barriers experienced by energy planners within Sierra Leone [13,14], it is not without limitations. Producing accurate demand forecasting is challenging due to the role of unpredictable trends and external factors which can influence energy demand and consumption [22]. Sierra Leone’s electricity sector has historically been impacted by external shocks such as COVID-19 and the Russia–Ukraine war, and it is likely to be impacted further in the future [9]. However, it is challenging to adequately represent and integrate such unpredictable external shocks into demand projections. Additionally, the socio-economic and technical data collected for this study often varied greatly depending on the sources used, with the ability to alter the projections produced depending on the sources selected. Whilst the ‘best-available’ data were identified and included within this study, data reliability

and consistency remains a significant barrier to accurate energy modelling and planning within Sierra Leone. Additionally, the MAED's simplified methodology lacks the ability to produce a detailed representation of the energy sector. Electricity is inputted and modelled as a single fuel, not allowing for variations across technologies, such as efficiencies and operating characteristics, which contribute to the overall electricity mix.

5.3. Conclusions

In this study, a quantitative simulation-modelling approach is applied to produce insights into possible demand forecasts under different socio-economic and technical parameters. A starter set of best-available economic, demographic, and technical data is collated and applied to the MAED-D software to provide an accessible, open-sourced capacity-building tool to be built upon in further work. Three modelled scenarios of baseline-, high-, and low-demand are created to produce demand projections from 2023 to 2050. This study shows that across the three modelled scenarios, demand forecasts by 2050 vary by 6.7 PJ, from 5.53 PJ to 12.23 PJ. This research study provides an insight into the possible applications of data to produce demand scenarios, and the impact that various socio-economic and technical parameters could have on the demand forecasts produced. It also overcomes barriers to energy planning in developing countries through collating a set of 'best-available' socio-economic and technical data for Sierra Leone. A baseline model and two further scenarios are developed using an open-source, accessible demand-modelling tool which in-country planners, academics, and policy makers can adopt to develop and integrate national energy development processes going forward.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/en17122878/s1>.

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Data Availability Statement: Data are contained within the article, Appendix A, and in the Supplementary Files. This work follows the U4RIA guidelines [57], which provide a set of high-level goals relating to conducting energy system analyses in countries. This paper was carried out involving stakeholders in the development of models, assumptions, scenarios, and results (Ubuntu/Community). The authors ensure that all data, source code, and results can be easily found, accessed, downloaded, viewed (retrievability), and licensed for reuse (reusability), and that the modelling process can be repeated in an automatic way (repeatability). The authors provide complete metadata for reconstructing the modelling process (reconstructability), ensuring the transfer of data, assumptions, and results to other projects, analyses, and models (interoperability), and facilitating peer-review through transparency (auditability).

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Appendix A

Table A1. Demographic parameters for Sierra Leone [45,58–64].

Parameters	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Population (Million)	7.861												
Population Growth Rate (%)		2.13	2.09	2.06	2.06	2.06	2.06	2.06	2.06	2.06	2.07	2.07	2.07
Urban Population (%)	42	42	43	43	44	44	45	45	46	47	48	49	50
Person/Urban Population (cap)	5	5	5	5	5	5	5	5	5	5	5	5	5
Person/Rural Population (cap)	6	6	6	6	6	6	6	6	6	6	6	6	6
Potential Labour Force (%)	72	72	72	72	72	72	72	72	72	72	72	72	72
Participating Labour Force (%)	54	54	54	54	54	54	54	54	54	54	54	54	54

Table A2. GDP parameters for Sierra Leone [6,10,49,59,65].

Parameters	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
GDP (USD Billion)	4.09										
GDP Growth Rate (%)		5.25	−2	4.1	3.98	2.75	4.74	5.19	4.55	4.47	4.6
Sectoral Shares of GDP (%)											
Industry (Total)	63.3	63.3	63.3	63.3	63.3	63.3	63.3	63.3	63.3	63.3	63.3
- Agriculture	57.4	57.4	57.4	57.4	57.4	57.4	57.4	57.4	57.4	57.4	57.4
- Mining	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9	5.9
Services	36.7	36.7	36.7	36.7	36.7	36.7	36.7	36.7	36.7	36.7	36.7
Total	100	100	100	100	100	100	100	100	100	100	100

Table A3. Total electrification rates from 2018 to 2023 [26,52,53].

Electrification %	2018	2019	2020	2021	2022	2023	2025	2030	2035	2040	2045	2050
Urban	53.2	51.4	55	57	57	57	61.82	73.87	85.92	98	99	100
Rural	6.4	4.7	4.7	4.9	4.9	4.9	10.9	26	41	56	71	86

Table A4. Total energy demand per sector from 2018 to 2023 [26,66].

PJ	2018	2019	2020	2021	2022	2023
Industry (Total)	0.101	0.158	0.139	0.145	0.18	0.182
- Agriculture	0.0131	0.02046	0.01804	0.01888	0.0234	0.02366
- Mining	0.0879	0.13754	0.12096	0.12612	0.1566	0.15834
Services	0.35	0.546	0.481	0.504	0.624	0.631
Household (Total)	0.205	0.319	0.281	0.295	0.365	0.369
- Urban	0.19475	0.30305	0.26695	0.28025	0.34675	0.35055
- Rural	0.01025	0.01595	0.01405	0.01475	0.01825	0.01845
Total	0.655	1.023	0.902	0.944	1.17	1.183

Table A5. Energy intensities by sector per annum from 2018 to 2050.

Energy Intensities	2018	2019	2020	2021	2022	2023	2030	2040	2050
Agriculture (MJ/USD)	0.00558	0.00828	0.00745	0.00749	0.00893	0.00879	0.01118	0.01577	0.02224
Mining (MJ/USD)	0.36426	0.54154	0.48598	0.48675	0.58126	0.57199	0.72773	1.02653	1.44802
Services (MJ/USD)	0.23317	0.34561	0.31068	0.31271	0.37235	0.36645	0.36645	0.36645	0.36645
Urban Households (MJ/dw/yr)	4856.38	7658.52	6031.96	5986.96	7093.12	7026.12	8683.19	11,669.5	15,682.8
Rural Households (MJ/dw/yr)	2124.67	4408.15	3715.09	3665.49	4342.73	4301.70	4941.31	6023.43	7342.52

Table A6. GDP and population growth rate assumptions from 2023 to 2050 across the three modelled scenarios.

Scenarios	2023	2024	2025	2030	2035	2040	2045	2050
GDP Growth Rate (%)								
Baseline	2.75	4.74	5.19	4.6	4.6	4.6	4.6	4.6
High	2.75	4.74	5.19	5.19	5.19	5.19	5.19	5.19
Low	2.75	2.75	2.75	2.75	2.75	2.75	2.75	2.75
Population Growth Rate (%)								
Baseline	2.06	2.06	2.06	2.06	2.06	2.06	2.06	2.06
High	2.06	2.54	2.54	2.54	2.54	2.54	2.54	2.54
Low	2.06	2.03	2	1.85	1.7	1.55	1.4	1.25

Table A7. Percentage share of sectoral GDP from 2023 to 2050 in the high-demand scenario.

Sectoral Shares of GDP (%)	2023	2024	2025	2026	2027	2028	2029	2030	2040	2050
Industry (Total)	63.3	62.83	62.36	61.89	61.42	60.95	60.48	60	60	60
- Agriculture	57.4	54.91	52.42	49.93	47.44	44.95	42.46	40	40	40
- Mining	5.9	7.92	9.94	11.96	13.98	16	18.02	20	20	20
Services	36.7	37.17	37.64	38.11	38.58	39.05	39.52	40	40	40
Total	100	100	100	100	100	100	100	100	100	100

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